

AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ May 2019

Physics Paper II

Time: 2 hours 10 minutes Marks: 50

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

**I agree that this is my name and school.
Candidate's Signature**

RUBRIC

2. There are THIRTEEN questions. Answer ALL questions. Questions 12 & 13 each offer TWO choices. Attempt any ONE choice from each.
3. When answering the questions:

Read each question carefully.
Use a black pointer to write your answers. DO NOT write your answers in pencil.
Use a black pencil for diagrams. DO NOT use coloured pencils.
DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
Complete your answer in the allocated space only. DO NOT write outside the answer box.
4. The marks for the questions are shown in brackets ().
5. You may use a scientific calculator if you wish.

Q.1. (Total 3 Marks)

The consequence of special theory of relativity gives the following relation of an object moving with a speed 'v' approaching to the speed of light 'c', relativistic mass 'm' and rest mass 'm₀'.

$$m = \frac{m_0}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$$

By using dimensions, show that 'm' in the given expression approaches infinity ($m \rightarrow \infty$).

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Q.2.

(Total 4 Marks)

Consider the given situations having no acceleration.

Situation 1: A man pulls a spring balance attached to a fixed wall with a force of 10 N.

Situation 2: Two different ropes are attached to each end of a spring balance. Two men pull the spring balance by applying a force of 5 N at each end in opposite directions.

- i. Identify whether the reading on the spring balance in **situation 1** is greater than, less than or equal to the reading on the spring balance in **situation 2**?
- ii. Identify the condition of equilibrium based on your answer in part (i). Justify your answer.

Q.3.

(Total 4 Marks)

Prove that the rate of change of linear momentum of a body is equal to the net force acting on the body.

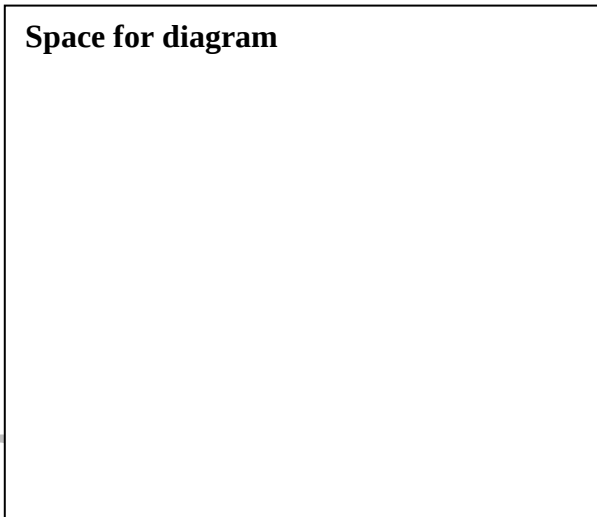
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Q.4. (Total 4 Marks)

Work done in the Earth's gravitational field is independent of the path followed by an object.

Explain the given statement. Draw a diagram to support your explanation.

Space for diagram



Q.5. (Total 3 Marks)

a. Define centripetal force and write its mathematical formula. (2 Marks)

b. Why must centripetal force be applied on an object to move it in a circular path? (1 Mark)

Q.6.

(Total 3 Marks)

Relate in THREE points the equation of continuity with the propulsion of a rocket.

Q.7.

(Total 4 Marks)

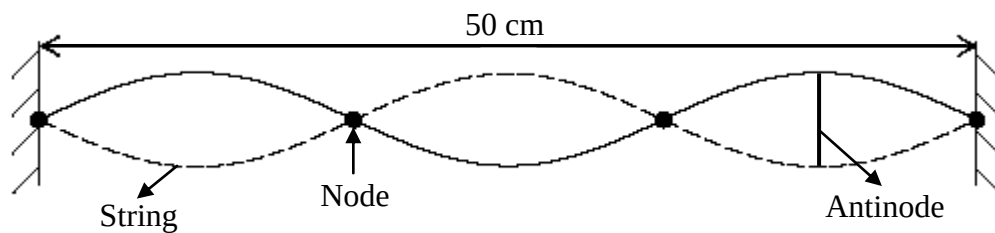
Calculate the frequency of a simple pendulum, if its length is 50 cm.

(**Note:** The acceleration due to gravity 'g' is 9.8 m/s^2 .)

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Q.8. (Total 4 Marks)

Standing waves are formed in an inextensible vibrating string as shown in the given diagram.



Find the wavelength of the standing wave.

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Q.9.

(Total 4 Marks)

Two ambulances 'A' and 'B' are moving with velocity of 20 m/s with normal siren frequency of 650 Hz. Ambulance 'A' moves towards a stationary listener while ambulance 'B' moves away from the stationary listener.

Calculate the frequency of siren heard by the stationary listener from each ambulance 'A' and 'B'.

(**Note:** The speed of sound in the air is 340 m/s.)

Q.10.

(Total 3 Marks)

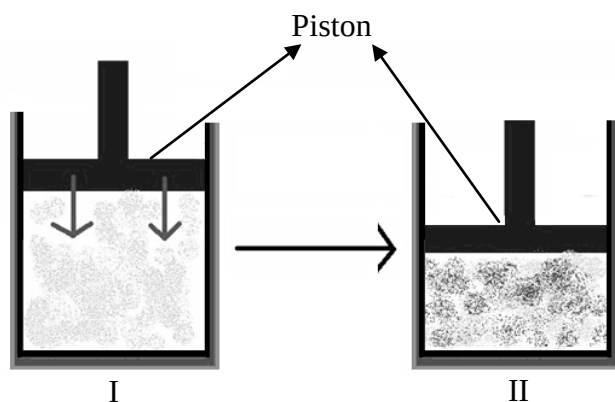
The polaroid sunglasses are better than ordinary sunglasses. Justify the given statement with reference to the glaring of light.

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Q.11.

(Total 4 Marks)

The given diagrams show a process in which the piston is compressed and there is no heat exchange.



- a. Identify the thermodynamic process illustrated in the given diagrams. (1 Mark)

- b. Discuss the relationship of the first law of thermodynamics for the process shown in the given diagrams when there is no heat transfer using mathematical equations. (2 Marks)

- c. In the illustrated process, will the temperature and internal energy raise or lower down? (1 Mark)

EITHER

- (Note:** The value of Universal Gravitational Constant is $6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$.)

OR

- (**Note:** The acceleration due to gravity ‘g’ is 9.8 m/s^2 .)

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Q.13.

(Total 5 Marks)

EITHER

- a. Oceanologists are discussing about the difference in the temperature of water at the surface of the ocean and the bottom of the ocean for operating a heat engine.

If they observed 20°C as high temperature and 4°C as low temperature, then calculate the efficiency of the heat engine. (5 Marks)

OR

- b. 100 J of heat is added in the system of a gas at constant pressure of $1.50 \times 10^2 \text{ Pa}$. If the system experiences a change in volume of 0.1 m^3 , then calculate the

i. work done by the gas. (3 Marks)

ii. internal energy change in the process. (2 Marks)

END OF PAPER

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